02244 Logic for Security
Information Flow
Week 11: Information Leakage

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About the Assignment 2

Many people have asked:

Do we have to implement the server?

Answer: NO! You should give a concept/design for a server:

- What security labels does it use (maybe user-definend)
- What API of commands does the server offer to the user
- What happens when these commands are executed
  - ★ This can be a short piece of (pseudo-) code or even plain English/Danish trying to be precise about what happens.
  - ★ E.g. for the uploaded code we perform the following information flow analysis (Approach, security lattice,...)
- Thus: how the server orchestrates the activities/security labels
- A security argument:
  - ★ what exactly are the security policies given by the labels/users?
  - ★ why can the server be sure that no security policies are ever violated? (e.g., use non-interference arguments)

Challenge

```
testrecord  = record [ subject : cpr  {shs:shs},
                       positive: bool {shs:shs} ]
publish_stat(db:array[testrecord:{⊥}]{⊥})
returns infections:int{⊥}

lookup_result(db:array[testrecord:{⊥}]{⊥},
              client:cpr{⊥})
returns result:bool{client:client}
```

`shs` is for sundhedsstyrelsen
Implement the functions `publish_stat` (giving total number of infections, i.e., positive testrecords) and `lookup_result` (giving the result of the specified client):

- What authority/declassifications do the functions need?
- Prove that this is indeed safe.

Challenge

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testrecord  = record[subject:cpr{shs:shs},positive:bool{shs:shs}]
publish_stat(db:array[testrecord:{⊥}]{⊥})
returns infections:int{⊥}{
  count:int {shs:shs} = 0;
  i:int{⊥} = 0;
  while (i<db.length){
    if (db[i].positive)
      count=count+1;

    i=i+1;

  }
  if_acts_for(publish_stat,shs){
    infections = declassify(count,{⊥});




  }
}
```

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returns infections:int{⊥}{
  count:int {shs:shs} = 0;
  i:int{⊥} = 0;
  while (i<db.length){
    if (db[i].positive)
      count=count+1; // Infoflow: db[i].positive,i,db.length --> count
                     // OK: {⊥},{shs:shs} <= {shs:shs}
    i=i+1; // Infoflow: i,db.length --> i
           // OK: {⊥} <= {⊥}
  }
  if_acts_for(publish_stat,shs){
    infections = declassify(count,{⊥});
    // Infoflow: count-->infections not allowed without declassification
    // declassif. count:{shs:shs} to {⊥} is allowed when acting as shs
    // Infoflow: declassify(count,{⊥}) -> infections
    // OK: {⊥} <= {⊥}
  }
}
```

Challenge

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testrecord  = record[subject:cpr{shs:shs},positive:bool{shs:shs}]
lookup_result(db:array[testrecord:{⊥}]{⊥},client:cpr{⊥})
returns result:bool{client:client}{
  int:i{⊥} = 0;
  tmp_result:bool{shs:shs} = false;
  while (i<db.length){
    if (db[i].subject=client){
      tmp_result=db[i].positive;




    }
    i = i+1; // similar as in publish_stat
  }
  if_acts_for(lookup_result,shs){
    result := declassify(tmp_result,{⊥});




  }
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returns result:bool{client:client}{
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  tmp_result:bool{shs:shs} = false;
  while (i<db.length){
    if (db[i].subject=client){
      tmp_result=db[i].positive;
      // Infoflow: i,db.length,db[i].subject,client,db[i].positive
      //           --> tmp_result
      // OK: {⊥},{shs:shs} <= {shs:shs}
    }
    i = i+1; // similar as in publish_stat
  }
  if_acts_for(lookup_result,shs){
    result := declassify(tmp_result,{⊥});
    // IF: tmp_result-->result not allowed without declassification
    // decl. tmp_result:{shs:shs} to {⊥} is allowed when acting as shs
    // Infoflow: declassify(tmp_result,{⊥}) -> result
    // OK: {⊥} <= {client:client}
  }
}
```

Monitoring Information Flow

So far, we have used information flow as a static analysis

- A compiler checks the information flow at compile time

What about an interpreter that checks it at run time?

- At any step monitoring that no illegal information is flowing, otherwise stopping execution.

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  - ★ before writing the result into x, check $\underline{x} \sqsubseteq \underline{t}$

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- No: Implicit flows! Example:
  `if y>0 then x := t` with $\underline{x} = L, \underline{y} = H$

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  `if y>0 then x := t` with $\underline{x} = L, \underline{y} = H$
  - ★ Suppose the interpreter remembers the context: all security labels of conditions under which the current command is (here $\underline{y} = H$)
  - ★ If writing into a lower variable (here $\underline{x} = L$), we stop execution.

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  - ★ That stopping is observable and thus leaks $y > 0$!

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  - ★ while computing t, compute the security label $\underline{t}$.
  - ★ before writing the result into x, check $\underline{x} \sqsubseteq \underline{t}$
  - ★ Is this sufficient?
- No: Implicit flows! Example:
  `if y>0 then x := t` with $\underline{x} = L, \underline{y} = H$
  - ★ Suppose the interpreter remembers the context: all security labels of conditions under which the current command is (here $\underline{y} = H$)
  - ★ If writing into a lower variable (here $\underline{x} = L$), we stop execution.
  - ★ That stopping is observable and thus leaks $y > 0$!
- So: NO, you need to make a bit of static analysis to identify implicit flows before they happen.

Timing Leaks

Consider the following naïve algorithm for $b^x \mod N$:

```
finished=false;
result=1;
while(x>0){
  result=result*b mod N;
  x=x-1;
}
finished=true
```

- Assume `finished` is level Low, while all other variables are High.
- This satisfies classical information flow.

Timing Leaks

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```

- Assume `finished` is level Low, while all other variables are High.
- This satisfies classical information flow.
- However, an intruder who can observe `finished` can estimate x — if they have a clock.
- This can be a problem in cryptographic algorithms, although the above example would never be used in crypto. (Why?)

Timing Leaks

Consider the following less naïve algorithm for $b^x \mod N$:

```
result=1;
while(x>1){
  if (x%2==0){x=x/2;}
  else{
    x=(x-1)/2;
    result=result*b % N;
  }
  b=b*b % N;
}
result=result*b % N;
```

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  }
  b=b*b % N;
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result=result*b % N;
```

- The runtime still depends on `x`
  - ★ position of the most significant set bit in `x`
  - ★ number of bits set in `x`

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  }
  b=b*b % N;
}
result=result*b % N;
```

- The runtime still depends on `x`
  - ★ position of the most significant set bit in `x`
  - ★ number of bits set in `x`
- It is not difficult to make this algorithm `constant time`:
  - ★ Ensure that also the `then` case performs a modular multiplication.
  - ★ Ensure that there are exactly η rounds of the `while` loop

    where η is the number of bits of `x`.

A Remote Timing Attack on RSA

As a more realistic example for timing leaks in cryptography, we consider now the paper

[1] D. Brumley and D. Boneh: *Remote timing attacks are practical*, Comput. Networks, 48(5), pp. 701–716, 2005.

- We skip some parts
- We make some simplifications
- You do not need to make any such analysis in your report!

Recall: RSA

Keygeneration:

- Choose bit-size η (recommended: $\eta \geq 2048$)
- Choose random `primes` p, q of size $\frac{\eta}{2}$.
- Let $N := p \cdot q$
- Choose random e of size η
- Compute $d := e^{-1} \mod (p-1)(q-1)$
- The resulting public key is (N, e)
  - ★ in OFMC for instance $\mathrm{pk}(B)$
- The resulting private key is (p, q, d)
  - ★ in OFMC for instance: $\mathrm{inv}(\mathrm{pk}(B))$

Encryption:

- $c := m^e \mod N$
  - ★ in OFMC for instance: $\{m\}_{\mathrm{pk}(B)}$

Decryption:

- $m := c^d \mod N$
  - ★ in OFMC: decryption handled as part of the Dolev-Yao rules

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- The resulting public key is (N, e)
  - ★ in OFMC for instance pk(B)
- The resulting private key is (p, q, d)
  - ★ in OFMC for instance: inv(pk(B))

Encryption: **to prevent guessing and multiplication attacks**

- $c := \boxed{\text{pad,m,rand}}^e \mod N$
  - ★ where *pad* is a fixed padding string, *rand* is a fresh random string

Decryption:

- $\boxed{\text{pad,m,rand}} := c^d \mod N$
  - ★ if pad is the fixed padding string, return m, otherwise error.

Montgomery Representation

- Computing the $\cdot \mod N$ operation is very expensive, but is needed up to 2η times during an exponentiation.
- Idea: with a different representation of numbers, we can reduce it to one single $\cdot \mod N$ operation!

Montgomery Representation (MR)

- ★ Let $R = 2^\eta$
- ★ Represent a number a as $\boxed{a \cdot R \mod N}$.
- ★ Note: while $a \cdot R$ has 2η bits, modulo N we have again η bits.
- ★ We just write $\boxed{a \cdot R}$ in the following for short.

Multiplication modulo N in MR

Multiplying two numbers a and b in MR:

$$\underbrace{\boxed{a \cdot R}}_{\eta \text{ bits}} \cdot \underbrace{\boxed{b \cdot R}}_{\eta \text{ bits}} \equiv_N \underbrace{\boxed{a \cdot b \cdot R \cdot R}}_{2\eta \text{ bits}}$$

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- Now the result must be divided by R to obtain $\boxed{a \cdot b \cdot R}$, i.e., the MR of $a \cdot b$.

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- Suppose:

$$\underbrace{\boxed{a \cdot b \cdot R \cdot R}}_{2\eta \text{ bits}} = \underbrace{\boxed{\text{XY}}}_{\eta \text{ bits}} \underbrace{\boxed{0\ldots0}}_{\eta \text{ bits}} \tag{1}$$

  ★ then the division by R would just be $\underbrace{\boxed{\text{XY}}}_{\eta \text{ bits}}$.

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  - ★ then the division by R would just be $\underbrace{\boxed{\text{XY}}}_{\eta \text{ bits}}$.

- But how to achieve (1)?

Multiplication modulo N in MR

- It is nearly certain that the lower η bits of $\boxed{a \cdot b \cdot R \cdot R}$ are not all zero.
- But note that $a \equiv_N a + N$, i.e. adding the modulus any number of times does not change the result modulo N.

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- For instance say $N = \langle 101 \rangle_2$

a	1	0	1	1	0	1	0	0
$+4N$	0	0	0	1	0	1	0	0
$\equiv_N a$	1	1	0	0	1	0	0	0

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- In this way we can achieve the lower half to be zero without changing the result modulo N.

Multiplication modulo N in MR

So we now have:

- Step 1:

$$\underbrace{\boxed{a \cdot R}}_{\eta \text{ bits}} \cdot \underbrace{\boxed{b \cdot R}}_{\eta \text{ bits}} \equiv_N \underbrace{\boxed{a \cdot b \cdot R \cdot R}}_{2\eta \text{ bits}}$$

- Step 2:

$$\underbrace{\boxed{a \cdot b \cdot R \cdot R + k \cdot N}}_{2\eta \text{ bits}} = \underbrace{\boxed{XY}}_{\eta \text{ bits}} \underbrace{\boxed{0...0}}_{\eta \text{ bits}}$$

- Step 3: $\boxed{a \cdot b \cdot R} \equiv_N \underbrace{\boxed{XY}}_{\eta \text{ bits}}$

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- But there is one small problem: XY may be larger than N (but not larger than $2N$)
  - ★ So Step 4: if $\mathrm{XY} > N$ subtract N from it!

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- But there is one small problem: XY may be larger than N (but not larger than $2N$)
  - ★ So Step 4: if $\mathrm{XY} > N$ subtract N from it!
- This last step is a timing leak!
  - ★ In the exponentation $c^d \mod N$, this is happening more often if c is close to N, and that can give a measurable difference!

Chinese Remainder Theorem

- So the exponentiation with MR can leak the modulus N.
- N in RSA is public, however, upon decryption, the factors $p \cdot q = N$ are known.
- Implementations usually exploit this using the Chinese Remainder Theorem:
  - ★ One can compute $c^d \mod N$ from the smaller problems
    - $c^{d \mod p} \mod p$
    - $c^{d \mod q} \mod q$
  - ★ This essentially makes it possible to obtain q by a timing attack, and thus the rest of the private key.
    - The paper uses also another timing leak in the multiplication.

The Timing Attack

Scenario: server running OpenSSL:

$$A \to B : \{PMS\}_{\mathsf{pk}(B)}, \ldots \quad (PMS \text{ is a fresh key})$$

The intruder can send an arbitrary value c and B will try to decrypt it. What to send as c?

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- Suppose the intruder already knows the a few of the most significant bits of q. Then the next bit can be obtained like this:

$$\begin{array}{lcccccccc} g & = & b_1 & b_2 & b_3 & 0 & 0 & 0 & 0 & 0 \\ g_{hi} & = & b_1 & b_2 & b_3 & 1 & 0 & 0 & 0 & 0 \end{array}$$

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- Measure the time for $c = g$ and $c = g_{hi}$ (under some transf.)

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- Timing reliably tells whether (a) or (b) is the case.
  - ★ This is a bit tricky, see original paper...

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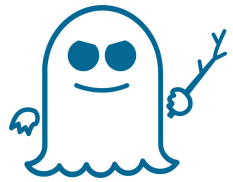
- Compute the next bit...

Counter-Measures

- Recommendations [1]:
  - ★ blinding: multiply c with random r before decryption (and divide result by r afterwards)
  - ★ constant time: ensure runtime does not depend on the data.
- Compiler Design with special forms of information flow can help here:
  - ★ Generate Code that does not have timing leaks.
  - ★ Compiler ensures that optimizations are not introducing time leaks.
  - ★ Target all popular processors.
  - ★ Verified implementation (no gap to mathematical model)
- Project Everest: https://project-everest.github.io
- **There is (almost) no excuse for using unverified crypto implementations!**

Spectre



Another spooky timing/side channel attack:

[3] Paul Kocher et al.: *Spectre attacks: exploiting speculative execution*, Commun. ACM, 63(7), pp. 93–101, 2020.

- Exploit combines two common performance optimizations of modern processors.
  - ★ Caching
  - ★ Speculative execution.
- Essentially: using speculative execution to leave "traces" in the cache and obtain these traces.
- Works on a wide variety of processors.

Speculative Execution

```
a=3;
b=Array[6];
c=f(a);
```

- Accessing `Array[6]` may be slow if not in cache.
- Instead of waiting, we could start computing `f(a)`.

Speculative Execution

```
a=3;
if (Array[6]>3)
  c=f(a);
else
  c=g(a);
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- If `Array[6]` takes time, we do not know for some time what is the next command.

Speculative Execution

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- Branch prediction: if we know `Array[6]>3` has been often true in the past, we could speculate it is going to be true again.
  - ★ Go ahead computing `f(a)`
  - ★ If `Array[6]>3` turns out to be true, we win.
  - ★ Otherwise, we have to revert the processor, but we did not loose anything.

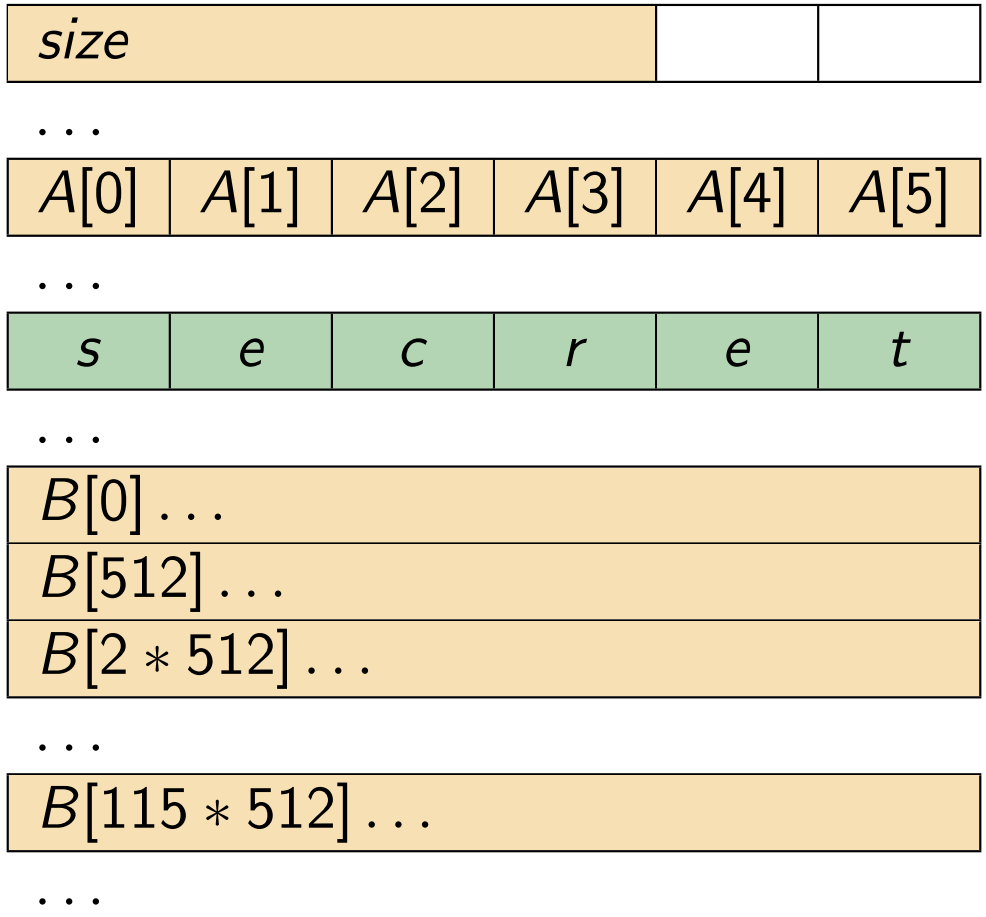
Spectre: Example

Victim code

```
if (y<size)
  temp &= B[A[y]*512]
```

Dramatis personae:

- `y`: chosen by the intruder.
- `A`: array of size `size`.
- The intruder is allowed to know arrays `A` and `B`.
- There is a *secret* in the application memory somewhere after `A`.



Spectre: Example

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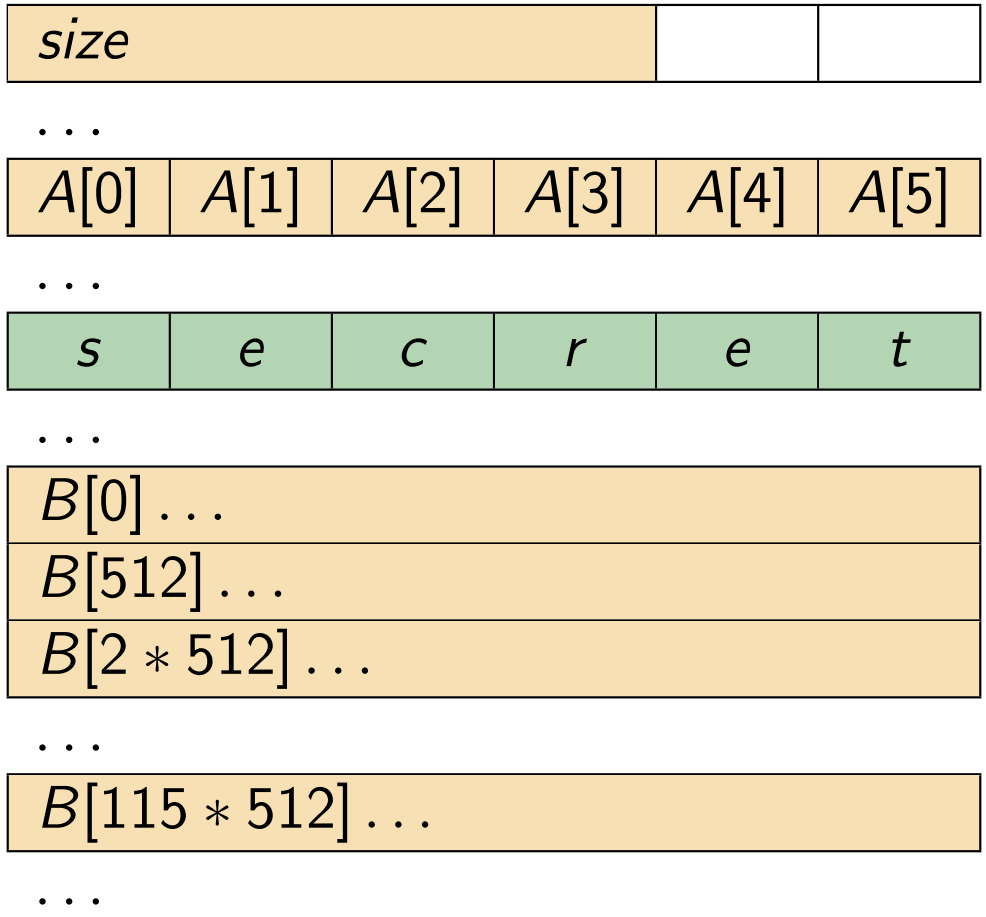
```

Suppose

- `size` is not in the cache.
- `A` is not in the cache.
- `B` is not in the cache.
- *secret* is in the cache.
- Branch prediction expects `y<size` to be true.



Spectre: Example

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```
if (y<size)
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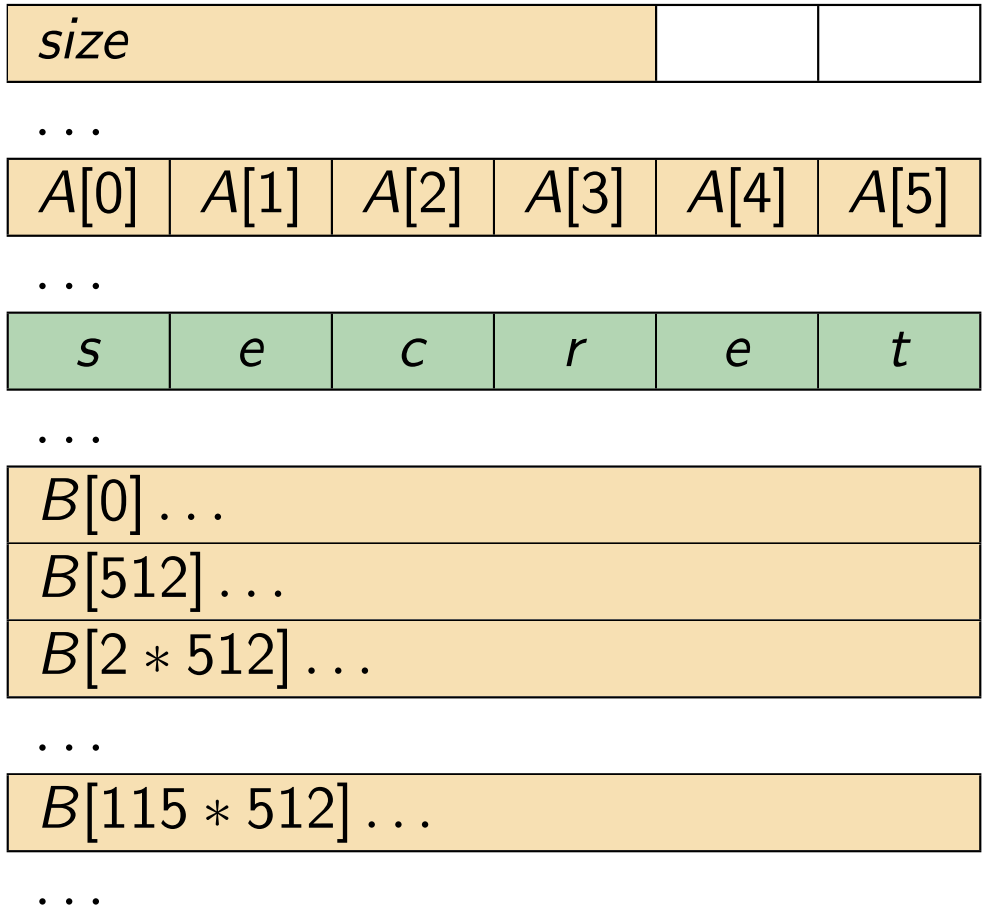
- The intruder guesses y, so that $A + y$ is the address of secret.
  - ★ So: `y>=size`!



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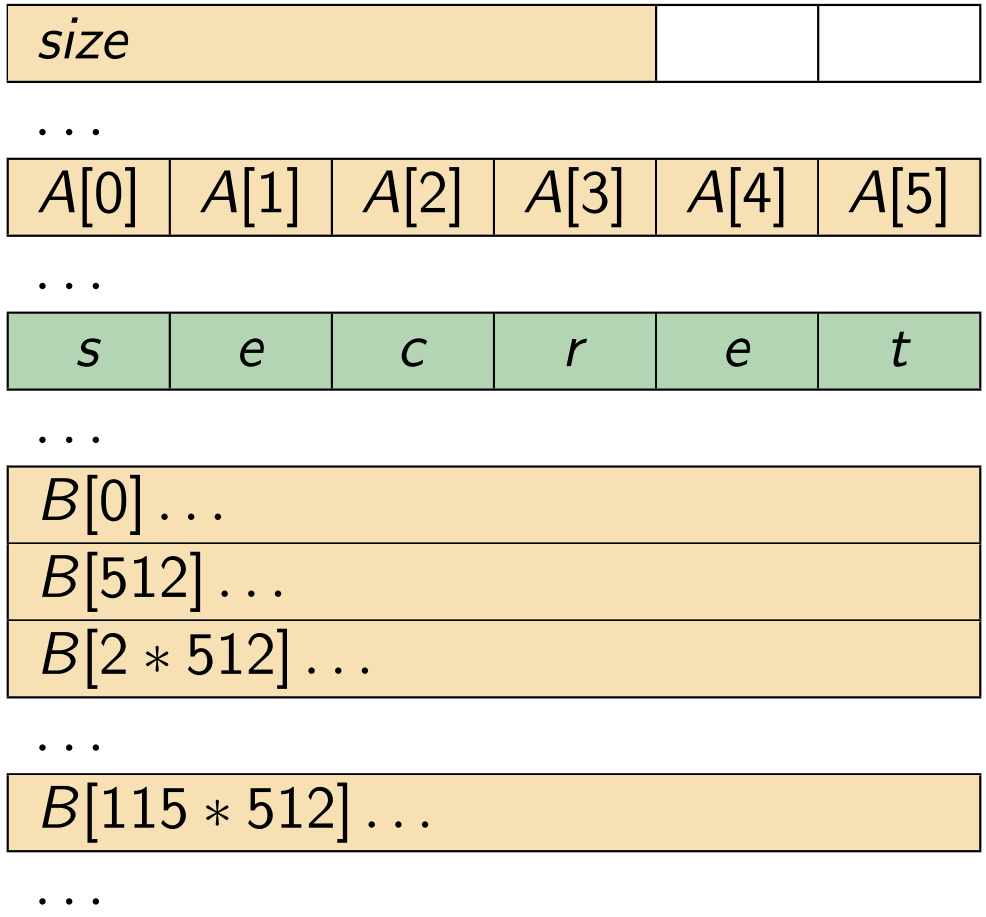
- Evaluate `y<size`:
  - ★ `size` not cached,
  - ★ branch prediction is positive



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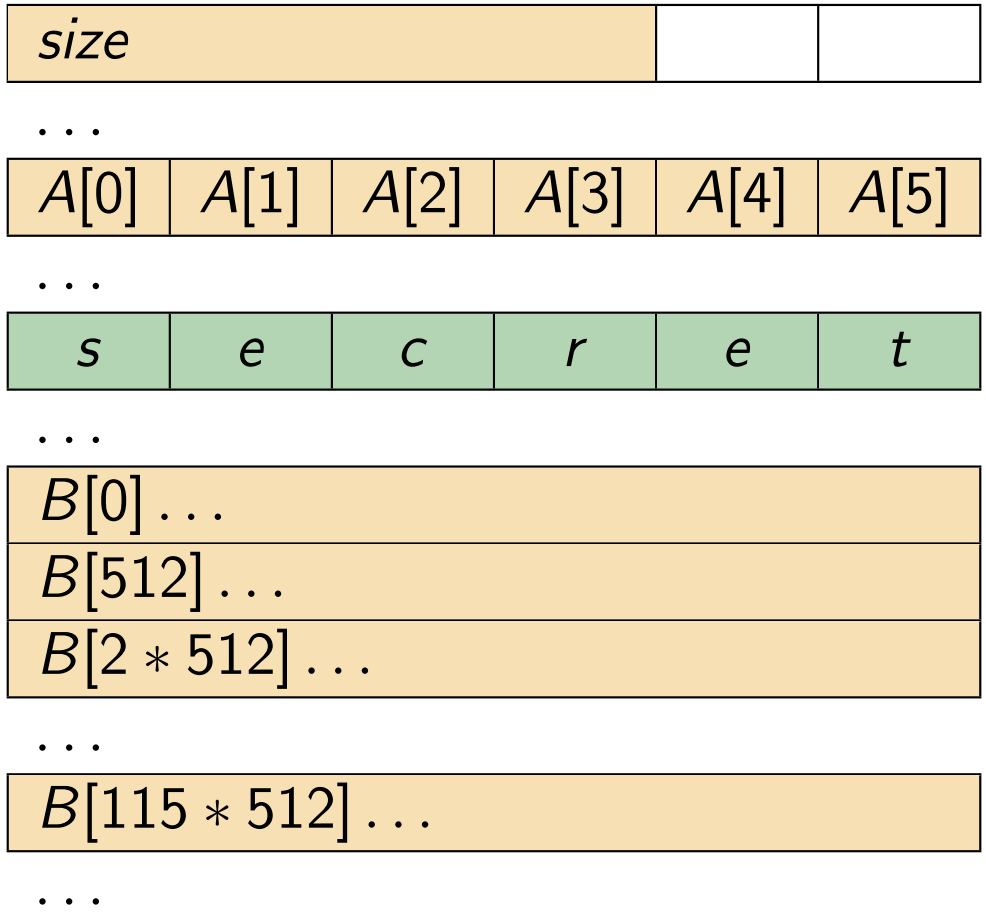

Thus start speculative execution of `B[A[y]*512]`.

Spectre: Example

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if (y<size)
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Speculative execution of `B[A[y]*512]`

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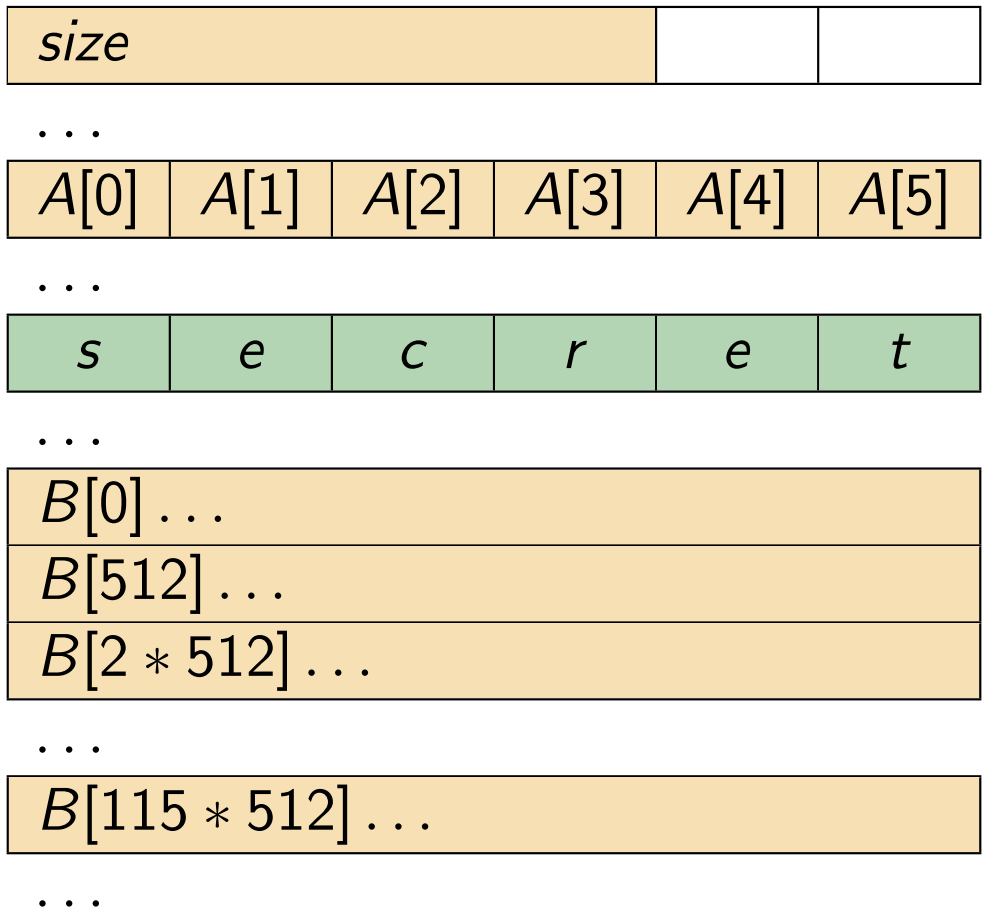

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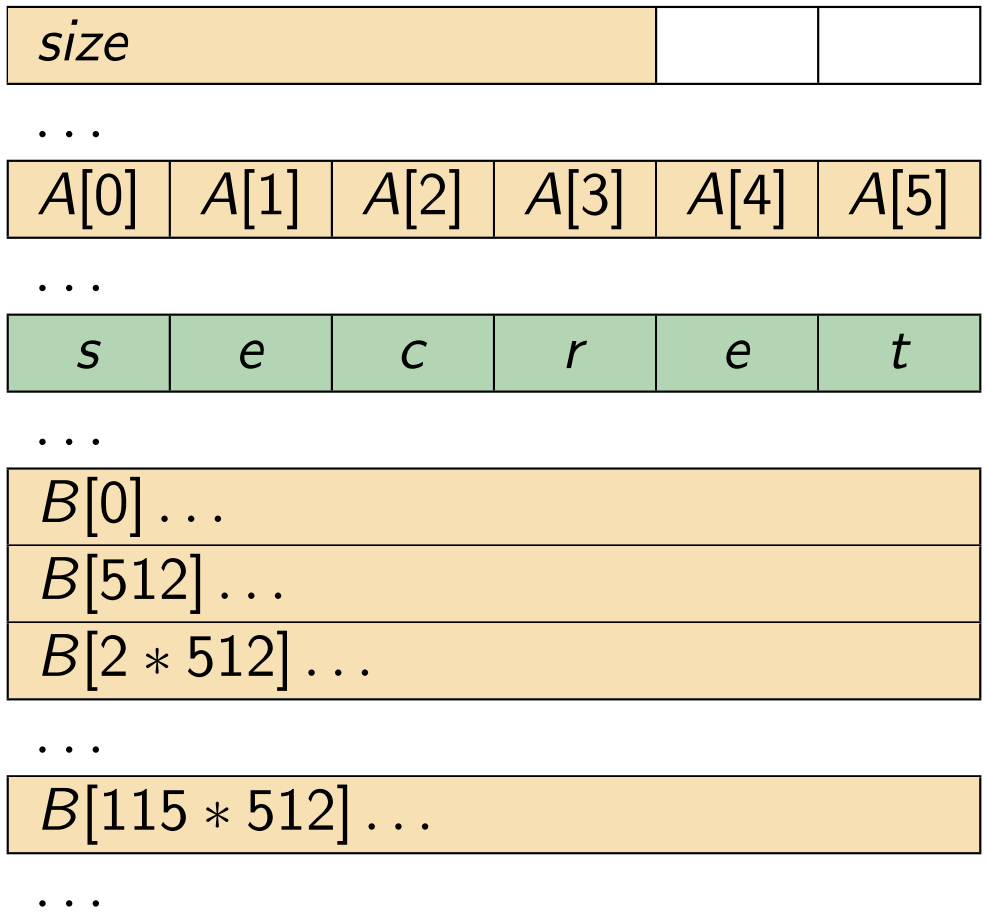

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Speculative execution of `B[A[y]*512]`

- $A + y$ is address of secret.
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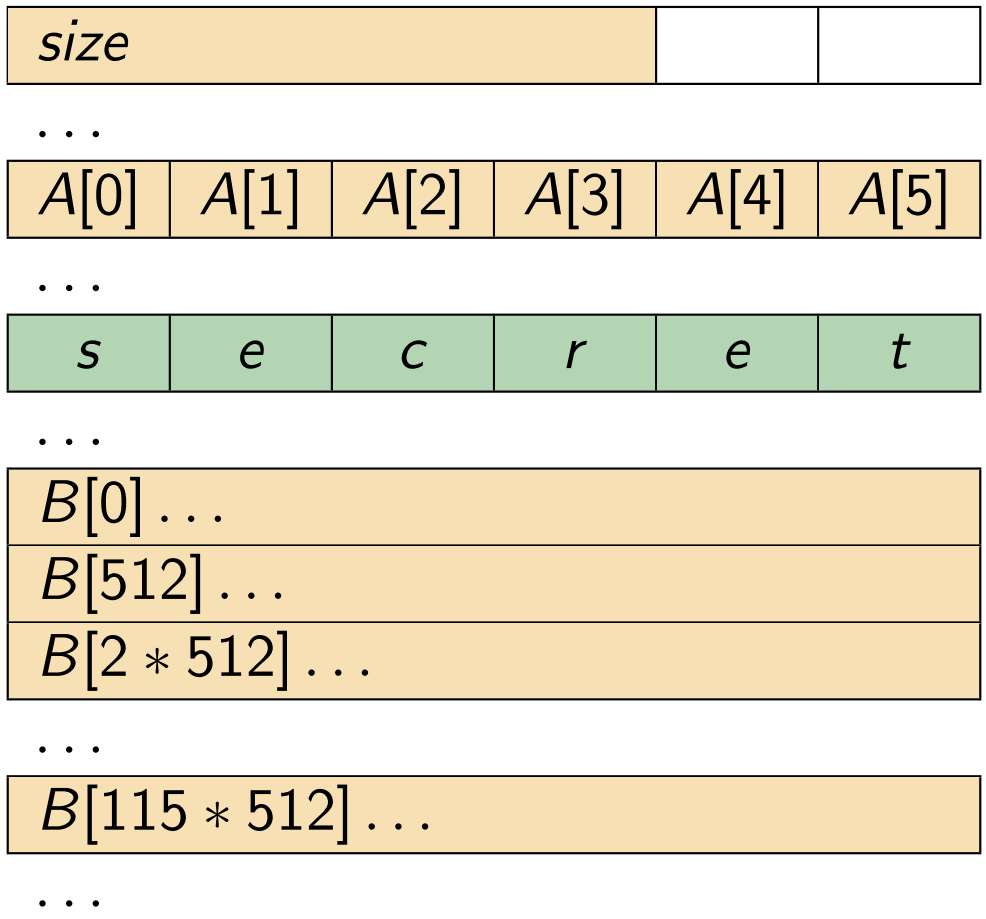
- So processor loads $B[A[y] * 512] = B[115 * 512]$



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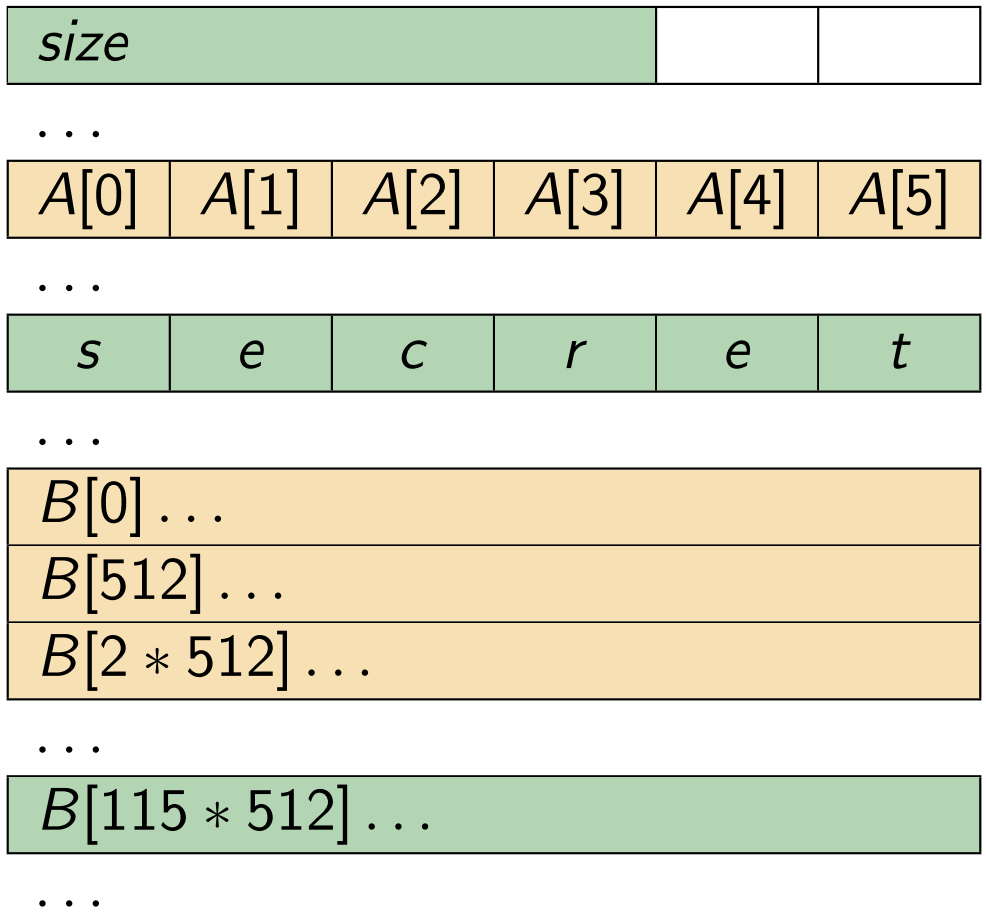
- $A + y$ is address of secret.
- So $A[y]$ is in the cache!
- So processor loads $B[A[y] * 512] = B[115 * 512]$
- ... it is now cached.



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- Thus, `y<size` can be computed to be false.
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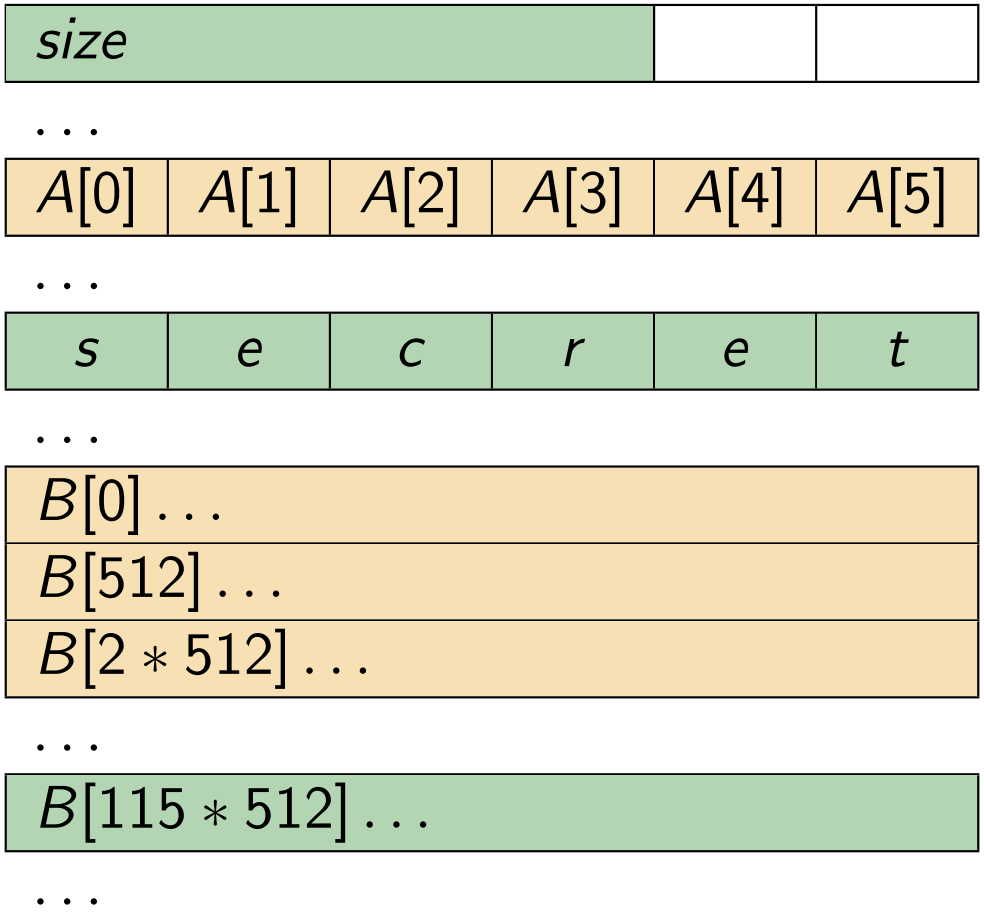

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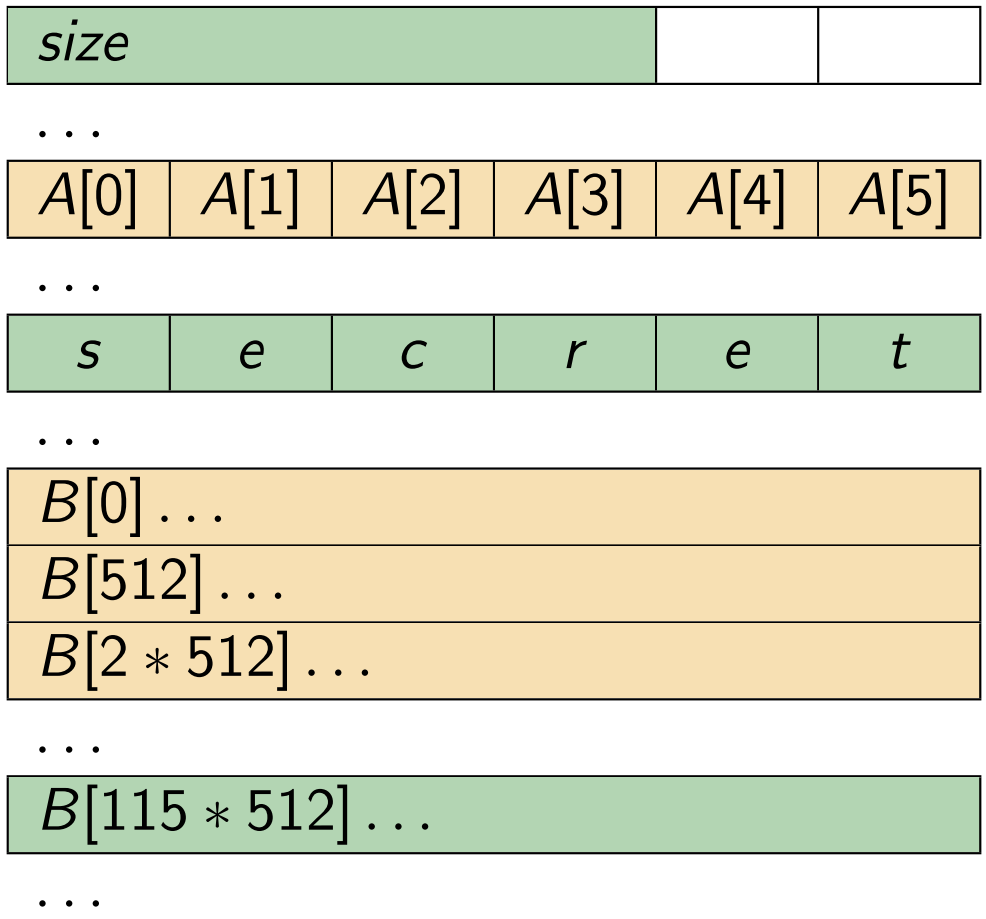

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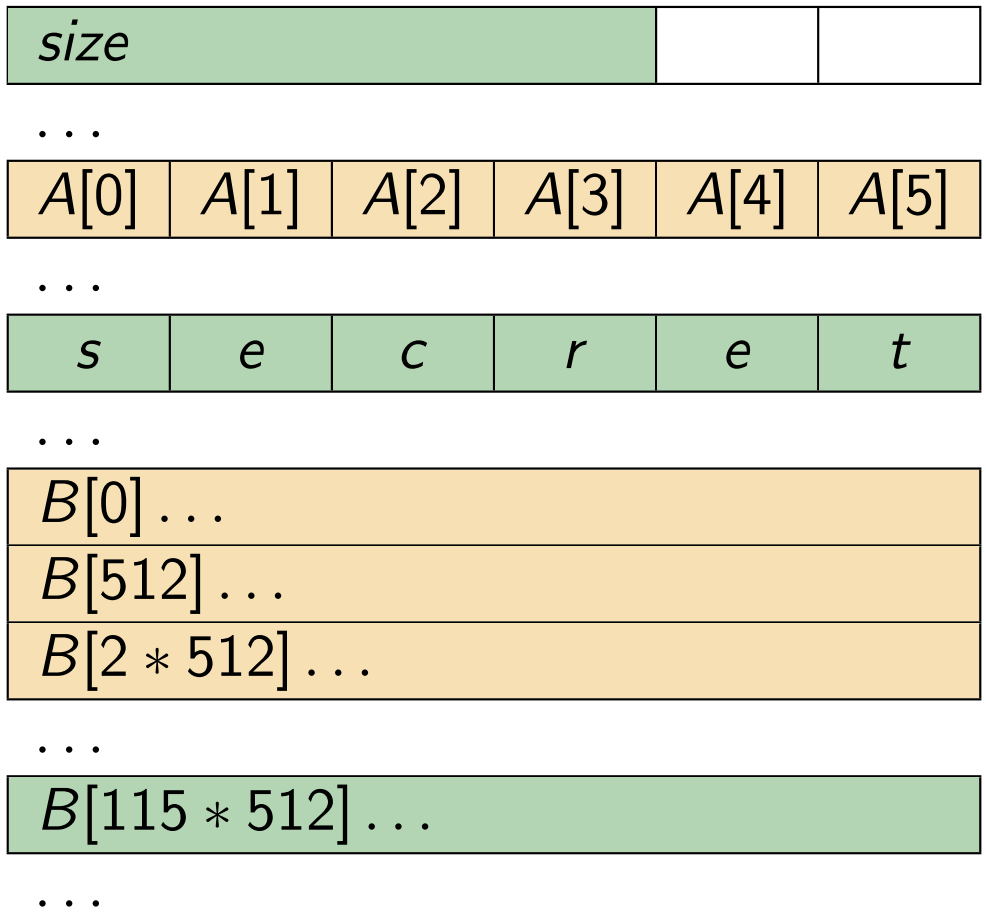

The intruder can now try to access `B[k*512]` for $k \in \{0..255\}$.

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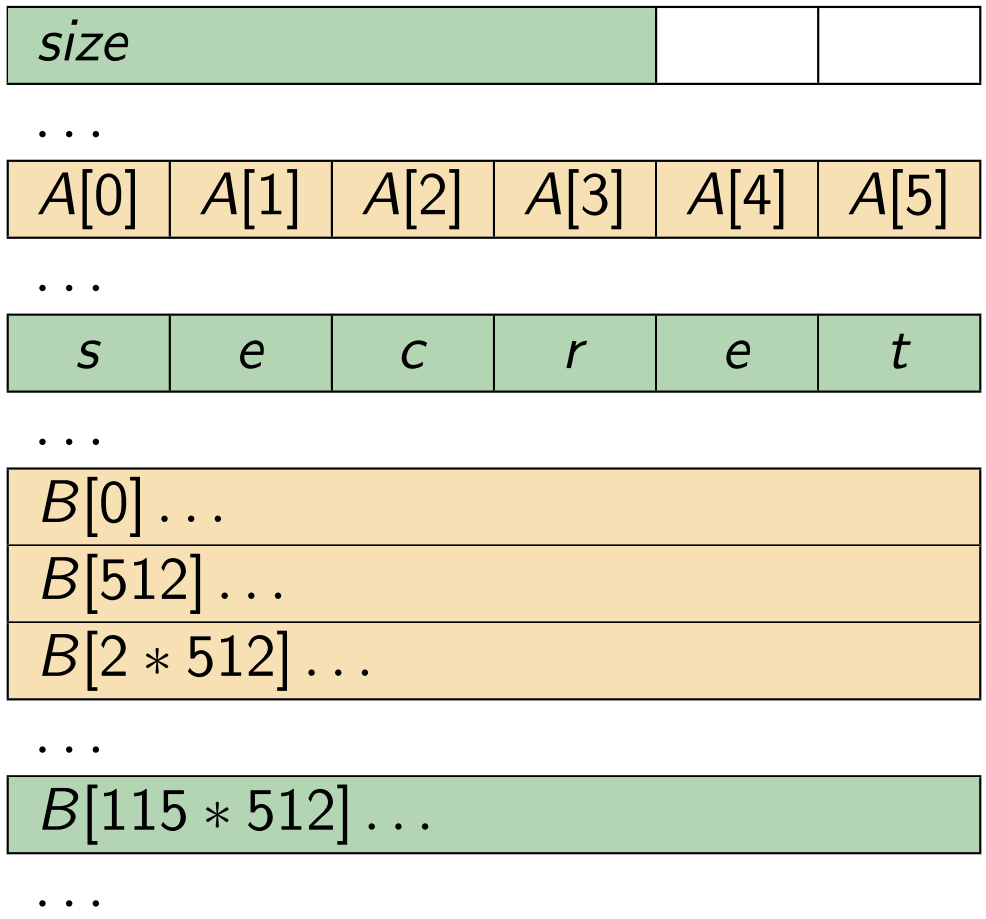

The intruder can now try to access `B[k*512]` for $k \in \{0..255\}$.
This is fast iff $k = 115$: he knows the first letter of the secret.

Speculative load hardening

The hardened code contains a few more operations:

```
        mov     size, %rax
        mov     y, %rbx
        mov     $0, $rdx
        cmp     %rbx, %rax
        jbe     END
        cmovbe  $-1, %rdx
        mov     A(%rbx), %rax
        shl     $9, %rax
        or      %rdx, %rax
        mov     B(%rax), %rax
        or      %rdx, %rax
        and     %rax, temp
END:    ...
```

Side Channel Analysis

- Side channels are usually leaks that the designers never have thought of.
- Many problems exist for years without getting noticed – how much else is out there?
- Very active research field:
  - ★ In order to prove a solution correct, one needs a precise model first.
  - ★ For instance Speculative non-interference and the Spectector tool. [2] Marco Guarnieri et al.: *Spectector: Principled Detection of Speculative Information Flows*, Security and Privacy, 2020.

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Spectector: Principled detection of speculative information flows.
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